Axiom Education

Name:	
Student ID:	

Vectors Basics [1.0N]

Specialist Mathematics 3/4 **Sample Worksheet**

Time: 1 hour 30 minutes

Mark
/40

- This Question and Answer Book consists of **11 questions**, with a total of **40 marks** available.

Approved materials

- Pens, pencils, highlighters, erasers, sharpeners, and rulers
- Protractors, set squares and aids for curve sketching

Materials supplied

- Question and Answer Book of 35 pages
- Formula Sheet
- Working space is provided throughout the book

Instructions

- Answer all questions in the Question and Answer Book.
- If additional space is needed to complete your answer, space is available at the end of this question book. Clearly write the number of the question that you are answering in this space.
- If you have any questions at any time during the examination, you must raise your hand and wait for a supervisor. Do **not** attempt to communicate, by any means, with any other student before or during the examination.
- Mobile phones, unauthorised electronic devices (including calculators or software), notes of any kind, blank sheets of paper, and/or correction fluid/tape are **not** permitted.
- Do not begin until instructed by your supervisor.

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Introduction to Vectors

Much of mathematics is built on numbers. Imagine you are watching a race at the Olympics. Consider the question 'how many laps does the race have?'.

You would likely respond with a singular number such as 'four'. But is this information enough in all scenarios?

Consider the question 'how long is this race?'. Is the answer 'four' still sufficient?

Here, a simple number is no longer enough. The question becomes four *what*? Clearly, we need to agree on some **unit** to describe the size of the quantity, such as a metre. These quantities are called **scalars**.

Suppose your friend is distracted and asks you 'do you know where Eliud Kipchoge is in the race right now?'. Now, even our improved answer is still insufficient. What do you think is missing?

A direction! '40 metres east of the finish line'. The scalar '40 metres' now has the necessary additional directional information 'east of the finish line'. Such quantities are called **vectors**.

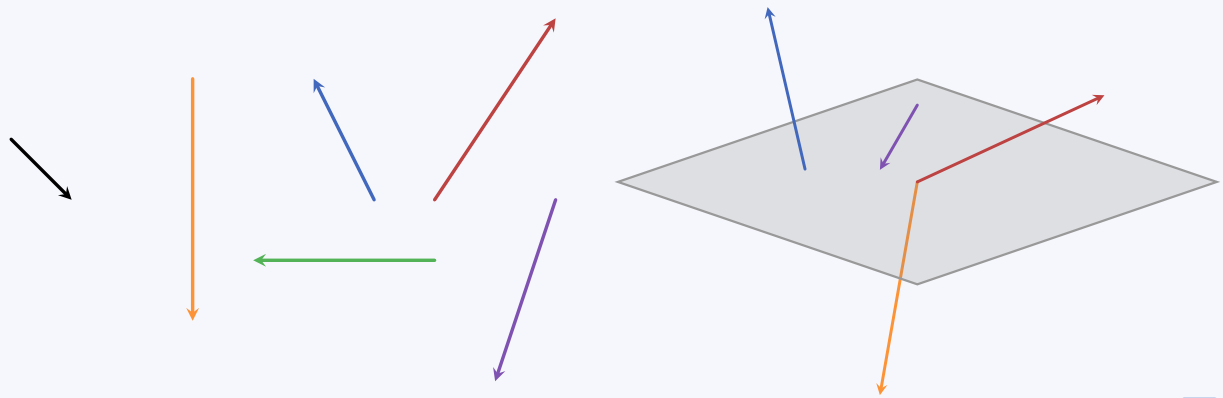
Vectors are the fundamental objects of linear algebra. In Specialist Mathematics, we'll deal with **real vectors**, which can be be defined in a geometric way.

Definition 1.1 Real Vectors

A *real vector* is an object with a **magnitude** (also called **norm** or **length**) and a **direction**.

In Specialist Mathematics, we refer to real vectors simply as *vectors*. Two-dimensional vectors can be drawn as arrows on a plane. Three-dimensional vectors can be drawn as arrows in space.

Example 1.1 Real Vectors



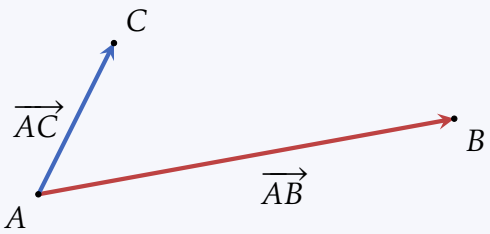
1.1 Vector Notation

Vectors are usually denoted in one of the following ways.

- **Arrow above letter** $\vec{u}, \vec{v}, \vec{x}, \vec{a}, \ldots$
  Mostly used in Physics!
- **Bold letter** $\mathbf{u}, \mathbf{v}, \mathbf{x}, \mathbf{a}, \ldots$
  This is most commonly used on computers, such as your CAS, since it's a little hard to hand-write bold letters.
- **Tilde below letter** $\underset{\sim}{u}, \underset{\sim}{v}, \underset{\sim}{x}, \underset{\sim}{a}, \ldots$
  This is the style used by VCAA, and **must** always be used in all your working.

A vector from point A to point B can also be denoted by $\overrightarrow{AB}$.

Example 1.2 Vectors Between Points



Theorem 1.1 Position Vectors

Given the **position vectors** of two points A and B, defined as $\underset{\sim}{a}$ and $\underset{\sim}{b}$ respectively, we have

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \underset{\sim}{b} - \underset{\sim}{a}$$

Note the order of subtraction is 'final – initial'. Think about why this is the case!

Challenge 1.1 Vector Addition

Consider the points $OABC$ in the Cartesian plane. Using a diagram, simplify the vector expression $\overrightarrow{BC} + \overrightarrow{AB} + \overrightarrow{OA} + \overrightarrow{CO}$.

1.2 Vector Components

Vectors are often represented in terms of several components. From our previous example, you might tell your friend that the best seat on the racetrack is '20 seats east, 10 seats north, 5 levels up'. Then the vector extending from your seat to the best seat would have three spatial components.

- 20 seats east
- 10 seats north
- 5 levels up

These three spacial components can combine to form the vector $\underset{\sim}{v}$.

Algebraically, we can represent this vector as $\underset{\sim}{v} = 20 \text{ seats east} + 10 \text{ seats north} + 5 \text{ levels up}$.

Each component of $\underset{\sim}{v}$ has a **vector** and **scalar** part.

- **Scalar Components:** Let $a = 20$ seats, $b = 10$ seats and $c = 5$ levels.
- **Vector Components:** Let $\underset{\sim}{e}_1$ be a unit vector pointing east, $\underset{\sim}{e}_2$ be a unit vector pointing north and $\underset{\sim}{e}_3$ be a unit vector pointing up.

Then the vector in terms of its scalar and vector components can be written as

$$\underset{\sim}{v} = a\underset{\sim}{e}_1 + b\underset{\sim}{e}_2 + c\underset{\sim}{e}_3$$

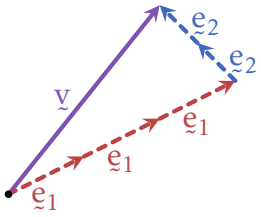
Notice how in this context, instead of choosing a single unit such as 'metres', we choose several directional 'units' $\underset{\sim}{e}_1$, $\underset{\sim}{e}_2$ and $\underset{\sim}{e}_3$, called **basis** vectors. These act as the fundamental building blocks for describing every vector in our space[1].

[1] This explanation was inspired by Kolecki, JC 2002, An Introduction to Tensors for Students of Physics and Engineering.

Suppose we choose the following directions for $\underset{\sim}{e}_1$ and $\underset{\sim}{e}_2$ in two dimensions.



Any vector in the plane can now be described by how much of each basis vector it contains. Consider the vector $\underset{\sim}{v}$ shown below.



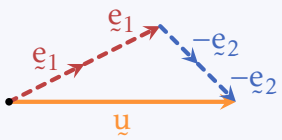
$$\underset{\sim}{v} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \begin{matrix} \leftarrow \text{How much of } \underset{\sim}{e}_1 \text{ is in } \underset{\sim}{v} \\ \leftarrow \text{How much of } \underset{\sim}{e}_2 \text{ is in } \underset{\sim}{v} \end{matrix}$$

From the diagram, we see that constructing $\underset{\sim}{v}$ requires three copies of $\underset{\sim}{e}_1$ and two copies of $\underset{\sim}{e}_2$. Algebraically, $\underset{\sim}{v} = 3\underset{\sim}{e}_1 + 2\underset{\sim}{e}_2$.

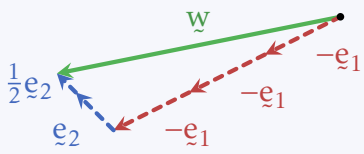
We can record this by writing $\underset{\sim}{v}$ as a 2×1 matrix $\underset{\sim}{v} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$, known as a **column vector**.

Example 1.3 Construction of Vectors Using Basis Vectors

Two more vectors represented as sums of the basis vectors $\underset{\sim}{e}_1$ and $\underset{\sim}{e}_2$ are shown below.



$$\underset{\sim}{u} = 2\underset{\sim}{e}_1 - 2\underset{\sim}{e}_2 \longrightarrow \underset{\sim}{u} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$



$$\underset{\sim}{w} = -3\underset{\sim}{e}_1 + 1.5\underset{\sim}{e}_2 \longrightarrow \underset{\sim}{w} = \begin{bmatrix} -3 \\ 1.5 \end{bmatrix}$$

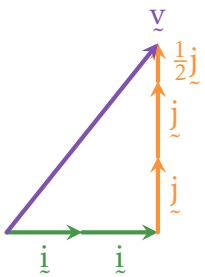
The entries of a column vector are called the **coordinates** (or **components**) of the vector, and they record how much of each basis vector is required to construct it.

Just like how some people use inches instead of centimetres, there is no 'correct choice' of basis vectors. The set of basis vectors used to define the coordinates of vectors is called a **coordinate system**, and a vector's coordinates depend on this choice.

For example, instead of using $\underset{\sim}{e}_1$ and $\underset{\sim}{e}_2$, we could choose a different pair of basis vectors, such as $\underset{\sim}{i}$ and $\underset{\sim}{j}$.



In this new coordinate system, the vector $\underset{\sim}{v}$ from earlier can be rewritten as

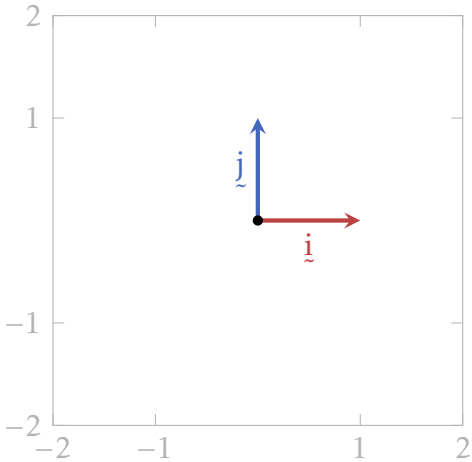


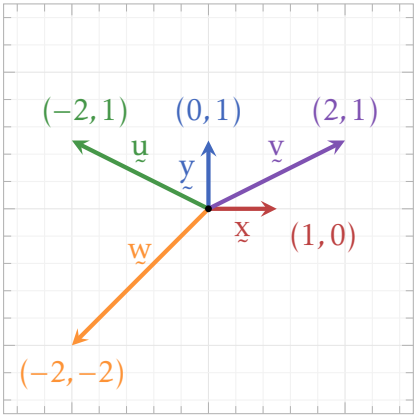
$$\underset{\sim}{v} = 2\underset{\sim}{i} + 2.5\underset{\sim}{j} \longrightarrow \underset{\sim}{v} = \begin{bmatrix} 2 \\ 2.5 \end{bmatrix}$$

Note 1.1 Vector Representation

A vector and its column representation are separate things! A vector is the same regardless of how it is represented. The exact way it is represented depends on the chosen coordinate system.

The space of two-dimensional vectors can be represented in a Cartesian coordinate system, with the origin of all vectors at the point $O = (0, 0)$. By convention, we choose $\underset{\sim}{i}$, a unit vector in the positive x-direction, and $\underset{\sim}{j}$, a unit vector in the positive y-direction as the standard basis vectors.





In three dimensions, the standard basis vectors are $\underset{\sim}{i}$, $\underset{\sim}{j}$ and $\underset{\sim}{k}$, with $\underset{\sim}{k}$ being a unit vector in the positive z-direction.

Note 1.2 VCAA Convention

VCAA does not use column vector notation. You can use column vectors during working, but it is best to present final answers in terms of the basis components $\underset{\sim}{i}$, $\underset{\sim}{j}$ and $\underset{\sim}{k}$.

Question 1.1 (2 marks)

Let $\underset{\sim}{a} = 2\underset{\sim}{i} + 3\underset{\sim}{j}$ and $\underset{\sim}{b} = -\underset{\sim}{i} + 3\underset{\sim}{j} - 2\underset{\sim}{k}$.

a) Represent $\underset{\sim}{a}$ and $\underset{\sim}{b}$ as column vectors. **1 mark**

b) Find $\underset{\sim}{a} - 2\underset{\sim}{b}$ **1 mark**

Question 1.2 (2 marks)

Simplify the expression $\overrightarrow{PQ} + \overrightarrow{RT} - \overrightarrow{RS} + \overrightarrow{QS}$.

Basic Vector Operations

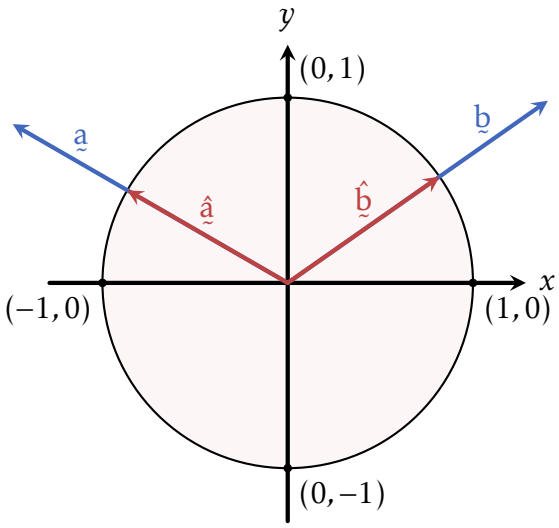
Now that we know what vectors are, we may want to define some operations. Given that a vector is a combination of a magnitude and direction, it would be useful to isolate these components.

Definition 2.1 Norm

The **norm** (or magnitude or length) of a vector $\underset{\sim}{v} = a\underset{\sim}{i} + b\underset{\sim}{j} + c\underset{\sim}{k}$ is given by

$$|\underset{\sim}{v}| = \sqrt{a^2 + b^2 + c^2}$$

This is based on the Pythagorean Theorem for both two and three dimensions.



Definition 2.2 Unit Vector

A **unit vector** is a vector with norm 1. To turn an arbitrary vector $\underset{\sim}{v}$ into a **unit vector** $\hat{\underset{\sim}{v}}$, the following formula can be used.

$$\hat{\underset{\sim}{v}} = \frac{1}{|\underset{\sim}{v}|}\underset{\sim}{v}$$

Challenge 2.1 Unit Vector Scaling

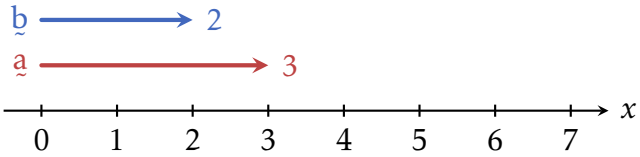
Consider the vector $\underset{\sim}{v} = 2\underset{\sim}{i} + \underset{\sim}{j}$. Construct a vector of length 3 that points in the direction of $\underset{\sim}{v}$.

2.1 Dot Product

We already have a notion of what it means to add or subtract two vectors. What about another fundamental operation, such as *multiplication*?

How can we think of multiplication of vectors? As vectors have two components, multiplying two vectors together isn't as easy as multiplying two numbers together.

Suppose we completely ignore the component of direction. Then all we have to focus on is the component of magnitude when multiplying two vectors together.



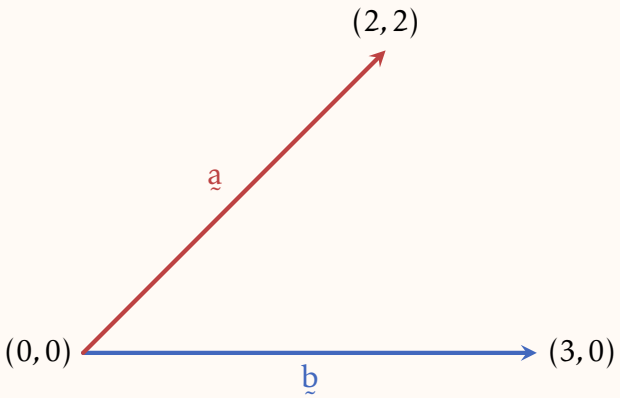
Hence, given that $\underset{\sim}{a}$ and $\underset{\sim}{b}$ are pointing in the same direction (i.e. the component of direction does not matter here), $\underset{\sim}{a} \cdot \underset{\sim}{b} = |\underset{\sim}{a}||\underset{\sim}{b}|$.

Note that this operation takes in two vectors and returns a scalar value. To differentiate this form of multiplication with another form that we will introduce later that produces a vector, we call this operation the **dot product**, appropriately named after the multiplication dot.

Perfect, now we have a version of multiplication for vectors! Can we apply this type of multiplication for vectors facing different directions?

Challenge 2.2 Vector Multiplication

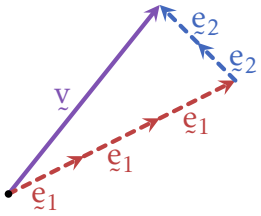
Given that multiplication takes two numbers to produce a new number, try to come up with an operation that takes in two vectors in any direction and produces a new number.



?

Using the formula you just came up with, we can conclude that $\underset{\sim}{a} \cdot \underset{\sim}{b} = |\underset{\sim}{a}||\underset{\sim}{b}|\cos(\theta)$. Can you work out why?

Recall this diagram from before, which is more interesting than we may have initially thought!



$$\underset{\sim}{v} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \begin{matrix} \leftarrow \text{How much of } \underset{\sim}{e}_1 \text{ is in } \underset{\sim}{v} \\ \leftarrow \text{How much of } \underset{\sim}{e}_2 \text{ is in } \underset{\sim}{v} \end{matrix}$$

On the left side, we have a vector represented as a geometric arrow in space. On the right side, we have represented that same vector as a list of numbers.

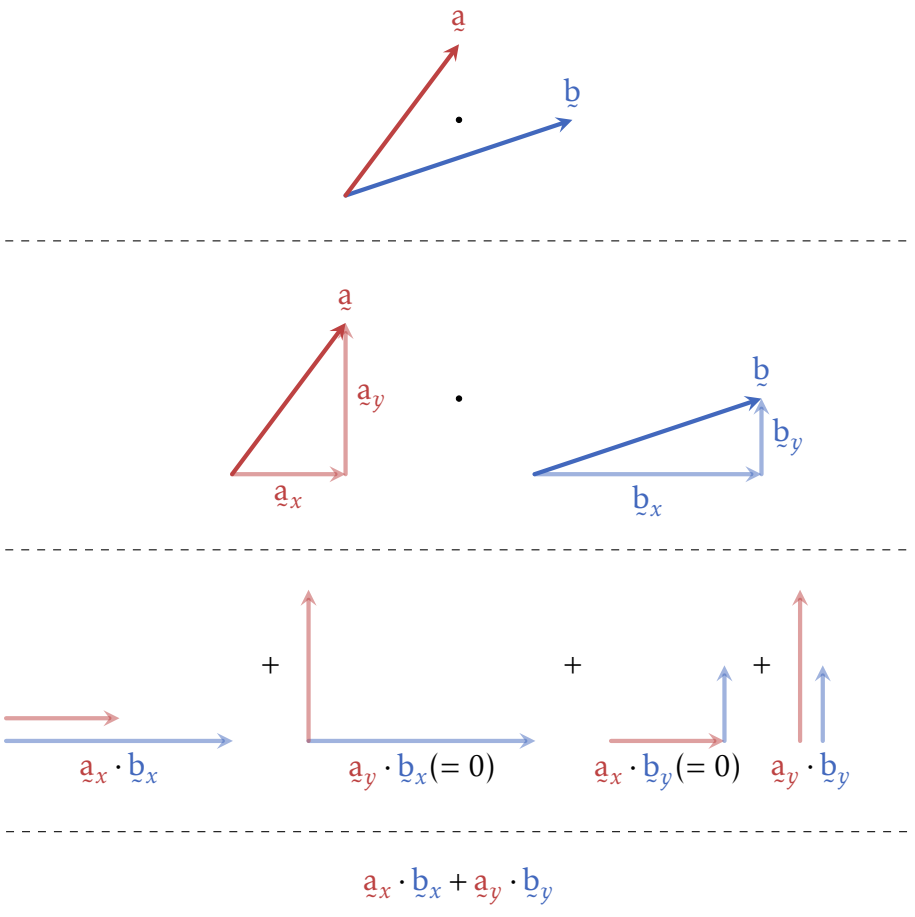
Often, physicists treat vectors solely as physical arrows, whereas computer scientists treat vectors solely as lists of numbers.

As mathematicians, we recognise that vectors are slightly more abstract. Both of these objects from two different worlds can provide us with interesting information about each other.

Our first introduction to the dot product involved a geometric approach using the geometric arrow. However, in Specialist Mathematics, your final answer will rarely be an arrow on your page, but rather a list of components and magnitudes, like below.

$$\underset{\sim}{v} = a\underset{\sim}{e}_1 + b\underset{\sim}{e}_2 + c\underset{\sim}{e}_3$$

As a result, though our geometric formula $\underset{\sim}{a} \cdot \underset{\sim}{b} = |\underset{\sim}{a}||\underset{\sim}{b}|\cos(\theta)$ is useful for our physical arrows, it becomes more difficult to compute with our lists of components and magnitudes.



$$\underset{\sim}{a}_x \cdot \underset{\sim}{b}_x + \underset{\sim}{a}_y \cdot \underset{\sim}{b}_y$$

Since the dot product is a form of multiplication for vectors, we can also interpret it as multiplying each of the rectangular components, as seen in the diagram above. This leaves us with two cases[2].

- The components are perpendicular, so the dot product is 0. Can you work out why?
- The components point in the same direction, meaning we can multiply the magnitudes like normal, as discussed here.

This leaves us with two final formulae for the dot product.

Definition 2.3 Dot Product

The **dot product** of two vectors $\underset{\sim}{a} = a_1\underset{\sim}{i} + a_2\underset{\sim}{j} + a_3\underset{\sim}{k}$ and $\underset{\sim}{b} = b_1\underset{\sim}{i} + b_2\underset{\sim}{j} + b_3\underset{\sim}{k}$ is given by

$$\underset{\sim}{a} \cdot \underset{\sim}{b} = |\underset{\sim}{a}||\underset{\sim}{b}|\cos(\theta) = a_1b_1 + a_2b_2 + a_3b_3$$

This interplay between the different worlds that vectors appear in hints at a much deeper relationship that veers outside the scope of Specialist Maths. If you are interested, we highly recommend watching 3Blue1Brown's Essence of Linear Algebra series, which introduces this topic in Chapter 9.

[2]This explanation was inspired by Vector Calculus: Understanding the Dot Product – BetterExplained 2026.

2.2 Resolutes

Suppose we fix a direction given by a vector $\underset{\sim}{b}$. Any other vector $\underset{\sim}{a}$ can be split into two parts.

- a part that is parallel to $\underset{\sim}{b}$.
- a part that is perpendicular to $\underset{\sim}{b}$.

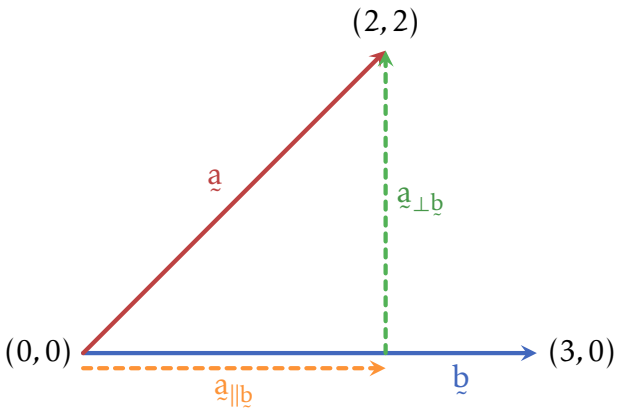
These are known as the **resolutes** or **projections** of $\underset{\sim}{a}$ onto $\underset{\sim}{b}$. This idea of a projection can be used for a variety of applications, such as in physics to find 'how much' of force is going in a particular direction, or in computing to render 3D visuals onto a 2D screen. Think of a projection of a vector like a 'shadow' onto another vector.

Since this is a useful concept, we can define a notation to represent this.

Definition 2.4 Vector Projections

Let $\underset{\sim}{a}$ and $\underset{\sim}{b}$ be two arbitrary vectors. Then

- the parallel (vector) resolute of $\underset{\sim}{a}$ onto $\underset{\sim}{b}$ is notated as $\underset{\sim}{a}_{\parallel \underset{\sim}{b}}$.
- the perpendicular resolute of $\underset{\sim}{a}$ onto $\underset{\sim}{b}$ is notated as $\underset{\sim}{a}_{\perp \underset{\sim}{b}}$.



Note 2.1 Alternative Notation

You may also see $\text{proj}_{\underset{\sim}{b}}\underset{\sim}{a}$ and $\text{oproj}_{\underset{\sim}{b}}\underset{\sim}{a}$ along your mathematical ventures! These mean the same thing as $\underset{\sim}{a}_{\parallel \underset{\sim}{b}}$ and $\underset{\sim}{a}_{\perp \underset{\sim}{b}}$ respectively.

2.2.1 The Scalar Resolute

As a reminder, vectors have both a direction and a magnitude. To find a formula for these resolutes, it may be helpful to focus on just the magnitude first.

Definition 2.5 Scalar Resolute

The magnitude of $\underset{\sim}{a}_{\|\underset{\sim}{b}}$ is called the **scalar resolute** of $\underset{\sim}{a}$ in the direction of $\underset{\sim}{b}$.

Challenge 2.3 Scalar Resolute

Suppose you have arbitrary vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$. Can you come up with a general formula that finds the scalar resolute of $\underset{\sim}{a}$ in the direction of $\underset{\sim}{b}$? Hint: $\underset{\sim}{a} \cdot \underset{\sim}{b} = \left|\underset{\sim}{a}_{\|\underset{\sim}{b}}\right| \cdot |\underset{\sim}{b}|$.

Challenge 2.4 Another Formula

An alternate formula for the **scalar resolute** is given by $\left|\underset{\sim}{a}_{\|\underset{\sim}{b}}\right| = \underset{\sim}{a} \cdot \hat{\underset{\sim}{b}}$. Show that this is equivalent to the formula derived in Challenge 2.3.

2.2.2 The Vector Resolute

Now that we can mathematically describe the magnitude of the parallel (vector) resolute, what about the direction?

Recall that we can scale any arbitrary vector $\underset{\sim}{v}$ into a unit vector $\hat{\underset{\sim}{v}}$. Hence, any arbitrary vector can be represented as a combination of a coefficient (magnitude) and a unit vector (direction).

Challenge 2.5 Vector Resolute

Given two arbitrary vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$, can you come up with a formula for $\underset{\sim}{a}_{\|\underset{\sim}{b}}$?

?

The formula you have derived above is not necessarily the easiest or most practical to compute! Often, mathematics textbooks provide the most computationally efficient formulae, leaving out the intuitive ones. It can be a useful exercise to bridge the gap between intuition and calculation to improve your mathematical versatility.

Challenge 2.6 Computational Formula

Given two arbitrary vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$, the computational formula for the vector resolute is $\underset{\sim}{a}_{\|\underset{\sim}{b}} = \dfrac{\underset{\sim}{a}\cdot\underset{\sim}{b}}{\underset{\sim}{b}\cdot\underset{\sim}{b}}\underset{\sim}{b}$. Show that this is equivalent to the formula derived in Challenge 2.5.

?

Question 2.1 (6 marks)

Let $\underset{\sim}{a} = 2\underset{\sim}{i} - \underset{\sim}{j} + 2\underset{\sim}{k}$, $\underset{\sim}{b} = -3\underset{\sim}{i} - 2\underset{\sim}{j} + \underset{\sim}{k}$ and $\underset{\sim}{c} = \underset{\sim}{i} - 2\underset{\sim}{j} + 2\underset{\sim}{k}$.

a) Find the cosine of the angle between $\underset{\sim}{a}$ and $\underset{\sim}{b}$. **2 marks**

b) Find the parallel resolute of $\underset{\sim}{b}$ in the direction of $\underset{\sim}{c}$. **2 marks**

c) Find a vector with magnitude $|\underset{\sim}{a}|$ that points in the same direction as $\underset{\sim}{c}$. **2 marks**

Question 2.2 (5 marks)

Consider the vectors $\underset{\sim}{u} = -3\underset{\sim}{i} + 2p\underset{\sim}{j}$ and $\underset{\sim}{v} = 6p\underset{\sim}{i} + 4\underset{\sim}{j} - 3\underset{\sim}{k}$, where $p \in \mathbb{R}$.

a) Given that $\underset{\sim}{u} + 2\underset{\sim}{v} = 7q\underset{\sim}{i} + 4q\underset{\sim}{j} - 6\underset{\sim}{k}$, where $q \in \mathbb{R}$, find the values of p and q. **3 marks**

b) Hence, find the angle between $\underset{\sim}{u}$ and $\underset{\sim}{v}$. **2 marks**

Question 2.3 (4 marks)

Let $\underset{\sim}{u} = 2\underset{\sim}{i} - \underset{\sim}{j} + a\underset{\sim}{k}$ and $\underset{\sim}{v} = b\underset{\sim}{i} + 5\underset{\sim}{j} - 4\underset{\sim}{k}$, where $a, b \in \mathbb{R}$.

a) Find the values of a and b such that $\underset{\sim}{u}$ and $\underset{\sim}{v}$ are parallel. **2 marks**

b) Hence, find the angle between $\underset{\sim}{u}$ and the positive direction of the x-axis. **2 marks**

Question 2.4 (3 marks)

Relative to a fixed origin, the points A, B and C are defined respectively by the position vectors $\underset{\sim}{a} = 2\underset{\sim}{i} + m\underset{\sim}{j} - \underset{\sim}{k}$, $\underset{\sim}{b} = \underset{\sim}{i} - 2\underset{\sim}{j} + 2\underset{\sim}{k}$ and $\underset{\sim}{c} = -\underset{\sim}{i} + \underset{\sim}{j} + \underset{\sim}{k}$, where $m \in \mathbb{R}$.

Given that $\angle ABC$ is $\frac{\pi}{3}$, find the value of m.

Linear Combinations

We have already discussed how we often choose a set of building blocks from which every other vector is constructed, our **basis** vectors!



An arbitrary vector $\underset{\sim}{v} = a\underset{\sim}{i} + b\underset{\sim}{j}$ in the Cartesian plane is a combination of two scaled vectors. This is called a **linear combination**!

Using linear combinations of $\underset{\sim}{i}$ and $\underset{\sim}{j}$, we can describe any vector that lies on the Cartesian plane.

Definition 3.1 Linear Combinations

An expression in the form

$$a_1\underset{\sim}{v}_1 + \cdots + a_n\underset{\sim}{v}_n$$

where $a, \ldots, a_n \in \mathbb{R}$ and $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_n$ are vectors is called a **linear combination**.

Challenge 3.1 Choosing a Basis

Can you choose any two vectors as your basis vectors? Can you think of any pairs of vectors that do not work?

Student Notes 3.1 Linear Combinations

3.1 Span

Let us consider one such exception where the vectors do not form a basis.



In the diagram above, both vectors point in the exact same direction. The green vector is simply a scaled version of the orange one. Algebraically,

$$\underset{\sim}{e}_2 = 2\underset{\sim}{e}_1$$

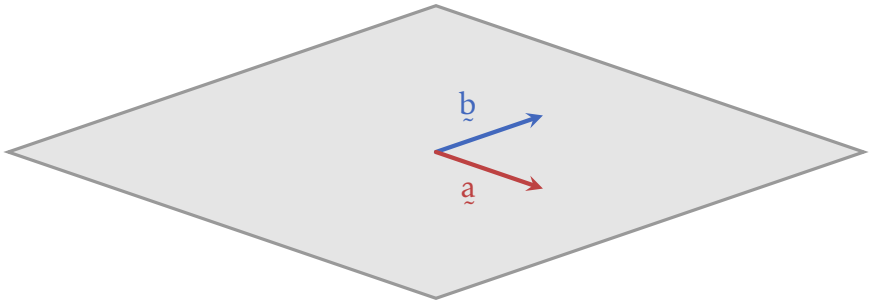
Challenge 3.2 Visualising Linear Combinations

Draw several linear combinations of $\underset{\sim}{e}_1$ and $\underset{\sim}{e}_2$, starting from an origin O. Do these points fill the plane, or are they restricted in some way?

Challenge 3.3 Three-Dimensional Linear Combinations

Suppose we have two vectors in three dimensions that are not parallel. Can we reach every point in three dimensions using these two vectors?

When combining two vectors that are not parallel, the points that we end up reaching form a plane. In two dimensions, this is the entire space, whereas in three dimensions, this is a subset of the space. Consider the diagram below.



The shaded region represents every point reachable from linear combinations of $\underset{\sim}{a}$ and $\underset{\sim}{b}$. No matter how we scale or add them together, we can never leave the plane itself!

Challenge 3.4 Reaching Three Dimensions

To reach every point in three dimensions, we can add a vector $\underset{\sim}{c}$ to form a new basis $\{\underset{\sim}{a}, \underset{\sim}{b}, \underset{\sim}{c}\}$. What properties does $\underset{\sim}{c}$ have?

Recall the concept of 'basis vectors'. Since it would be useful to describe the region in space that the basis vectors can reach, we define the **span** as exactly that!

Definition 3.2 Span

Let S be a set of vectors. The **span** of S, denoted $\text{span}(S)$ is the set of all linear combinations that can be formed using the vectors in S.

Note 3.1 Span In Specialist Maths

The **span** is not explicitly tested in Specialist Maths! However, it is still a useful concept to introduce and build intuition for later topics.

3.2 Linear Independence

Challenge 3.5 Redundancy

Consider the two-dimensional vectors $\underset{\sim}{e}_1$ and $\underset{\sim}{e}_2$ illustrated below.



What is the span of these two vectors? Does the span change if you add another vector? What about the three-dimensional case?

In the case where we add a third vector $\underset{\sim}{e}_3$ that lies in the span of $\underset{\sim}{e}_1$ and $\underset{\sim}{e}_2$, the span does not change. This is because this new vector can simply be represented using the other two vectors. In this fashion, the span of $\{\underset{\sim}{e}_1, \underset{\sim}{e}_2\}$ is equal to the span of $\{\underset{\sim}{e}_1, \underset{\sim}{e}_2, \underset{\sim}{e}_3\}$.

When we have a set of vectors and can remove one without reducing the span, we call this set of vectors **linearly dependent**. This is equivalent to saying that one of the vectors can be expressed as a linear combination of the others.

A set of vectors that is not linearly dependent is called **linearly independent**. Can you think of why?

Definition 3.3 Linear Independence

A set of vectors S is **linearly dependent** if at least one of the vectors can be written as a linear combination of the others. In the case of three vectors $\{\underset{\sim}{v}_1, \underset{\sim}{v}_2, \underset{\sim}{v}_3\}$, the expression

$$\underset{\sim}{v}_3 = a_1\underset{\sim}{v}_1 + a_2\underset{\sim}{v}_2$$

must have a solution, where $a_1, a_2 \in \mathbb{R}$.

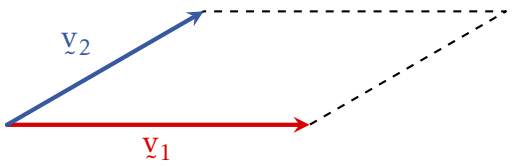
A set of vectors S that is not linearly dependent is called **linearly independent**.

Student Notes 3.2 More Linear Combinations

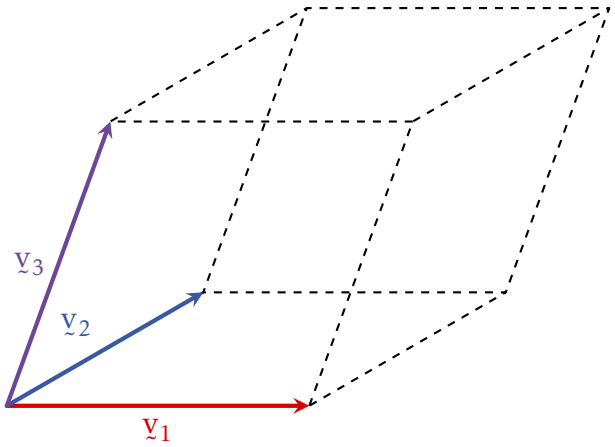
3.3 The Determinant

Let us consider another visualisation of linear independence!

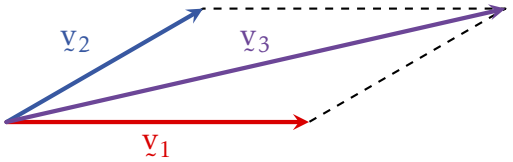
In two dimensions, when two vectors are placed tail-to-tail, they form a parallelogram.



In three dimensions, this idea extends naturally. When three vectors are placed tail-to-tail, they form a three-dimensional shape known as a parallelepiped. If the three vectors point in different directions, the parallelepiped has a non-zero volume. In this case, none of the vectors are redundant, so the of set vectors is **linearly independent**.



However, if one of the vectors lies in the plane formed by the other two, the parallelepiped collapses into a flat shape with zero volume. This means that one vector can be written as a linear combination of the others, so the set of vectors is **linearly dependent**.



Challenge 3.6 Linear Independence Conditions

From this new geometric approach, can come up with a rule to describe when a set of vectors in three dimensions is linearly independent?

?

We can calculate the volume of the parallelepiped using something called the determinant! This allows us to utilise the condition we derived in Challenge 3.6.

Definition 3.4 Parallelepiped Volume With Determinant

Given three vectors $\underset{\sim}{v}_1 = x_1\underset{\sim}{i} + y_1\underset{\sim}{j} + z_1\underset{\sim}{k}$, $\underset{\sim}{v}_2 = x_2\underset{\sim}{i} + y_2\underset{\sim}{j} + z_2\underset{\sim}{k}$ and $\underset{\sim}{v}_3 = x_3\underset{\sim}{i} + y_3\underset{\sim}{j} + z_3\underset{\sim}{k}$, we have their column representations $\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}$, $\begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$ and $\begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$ respectively.

Adjoining these column vectors allows us to produce a 3×3 matrix $\begin{bmatrix} x_1 & x_2 & x_2 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix}$

The determinant of this matrix is given by

$$\begin{vmatrix} x_1 & x_2 & x_2 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{vmatrix} = x_1 \begin{vmatrix} y_2 & y_3 \\ z_2 & z_3 \end{vmatrix} - x_2 \begin{vmatrix} y_1 & y_3 \\ z_1 & z_3 \end{vmatrix} + x_3 \begin{vmatrix} y_1 & y_2 \\ z_1 & z_2 \end{vmatrix}$$

$$= x_1(y_2z_3 - y_3z_2) - x_2(y_1z_3 - y_3z_1) + x_3(y_1z_2 - y_2z_1)$$

which evaluates to the signed volume of the parallelepiped formed by $\underset{\sim}{v}_1$, $\underset{\sim}{v}_2$ and $\underset{\sim}{v}_3$.

Example 3.1 Linear Independence Using Determinant

Let $\underset{\sim}{a} = \underset{\sim}{i} + 2\underset{\sim}{j} - \underset{\sim}{k}$, $\underset{\sim}{b} = -\underset{\sim}{j} + 2\underset{\sim}{k}$ and $\underset{\sim}{c} = 2\underset{\sim}{i} + \underset{\sim}{j} + \underset{\sim}{k}$. Show that $\{\underset{\sim}{a}, \underset{\sim}{b}, \underset{\sim}{c}\}$ is linearly independent.

Solution:

The column representations of $\underset{\sim}{a}$, $\underset{\sim}{b}$ and $\underset{\sim}{c}$ are $\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$, $\begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$ respectively.

Adjoin these column vectors together to form a matrix $\begin{bmatrix} 1 & 0 & 2 \\ 2 & -1 & 1 \\ -1 & 2 & 1 \end{bmatrix}$.

Then $\begin{vmatrix} 1 & 0 & 2 \\ 2 & -1 & 1 \\ -1 & 2 & 1 \end{vmatrix} = 1 \cdot \begin{vmatrix} -1 & 1 \\ 2 & 1 \end{vmatrix} - 0 \cdot \begin{vmatrix} 2 & 1 \\ -1 & 1 \end{vmatrix} + 2 \cdot \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} = (-1-2) - 0 + 2 \cdot (4-1) = 3 \neq 0$

Therefore $\{\underset{\sim}{a}, \underset{\sim}{b}, \underset{\sim}{c}\}$ is linearly independent.

Note 3.2 Rule for Computing Determinant

The formula that we use for the 3×3 determinant is known as **cofactor expansion**! The proof for this requires knowledge that is well beyond the scope of the course, so take it as face value for now. Feel free to go on an internet rabbit-hole if you're interested :)

Finally, here is a general guide of how you should approach different questions with linear independence.

Dimension	1 Vector	2 Vectors	3 Vectors	≥ 4 Vectors
2D	Lin. indep.	Check if parallel	Lin. dep.	Lin. dep.
3D	Lin. indep.	Check if parallel	Check determinant	Lin. dep.

Student Notes 3.3 Linear Independence

Question 3.1 (2 marks)
Consider the vectors $\underset{\sim}{a} = -2\underset{\sim}{i} + \underset{\sim}{j} + \underset{\sim}{k}$, $\underset{\sim}{b} = \underset{\sim}{i} - 2\underset{\sim}{j} + 3\underset{\sim}{k}$ and $\underset{\sim}{c} = m\underset{\sim}{i} + \underset{\sim}{j} - \underset{\sim}{k}$, where $m \in \mathbb{R}$.

a) Evaluate $\underset{\sim}{a} - 2\underset{\sim}{b}$. **1 mark**

b) Find a value of m such that $\{\underset{\sim}{a}, \underset{\sim}{b}, \underset{\sim}{c}\}$ is linearly dependent. **1 mark**

Question 3.2 (5 marks)
Consider the following vectors $\underset{\sim}{u} = 2\underset{\sim}{i} - \underset{\sim}{j} + 3\underset{\sim}{k}$, $\underset{\sim}{v} = 3\underset{\sim}{i} + 4\underset{\sim}{j} - \underset{\sim}{k}$ and $\underset{\sim}{w} = a\underset{\sim}{i} + \underset{\sim}{j} + 4\underset{\sim}{k}$, where $a \in \mathbb{R}$.

a) Find the value of a for which the vectors $\underset{\sim}{u}$, $\underset{\sim}{v}$ and $\underset{\sim}{w}$ are linearly dependent. **3 marks**

b) Hence, find the angle between $\underset{\sim}{u}$ and $\underset{\sim}{w}$. **2 marks**

Question 3.3 (4 marks)

Consider the three vectors $\underset{\sim}{a} = \underset{\sim}{i} - \underset{\sim}{j} + 2\underset{\sim}{k}$, $\underset{\sim}{b} = -2p\underset{\sim}{i} + \underset{\sim}{j} + 2\underset{\sim}{k}$ and $\underset{\sim}{c} = \left(p^2 - 3\right)\underset{\sim}{i} - \underset{\sim}{j} - 2\underset{\sim}{k}$, where $p \in \mathbb{R}$.

Find the values of p such that $\{\underset{\sim}{a}, \underset{\sim}{b}, \underset{\sim}{c}\}$ is linearly independent.

Geometric Proofs

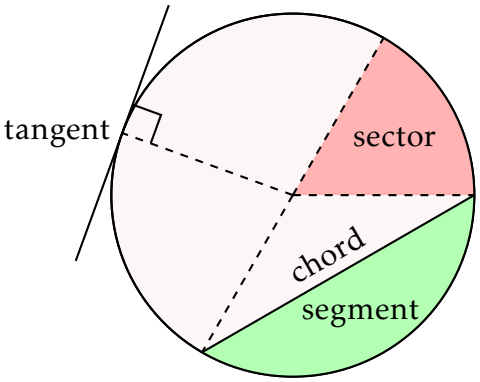
Throughout this worksheet, we have defined vectors as some geometric object with a magnitude and direction. We have also developed vectors algebraically, creating some useful tools to manipulate them.

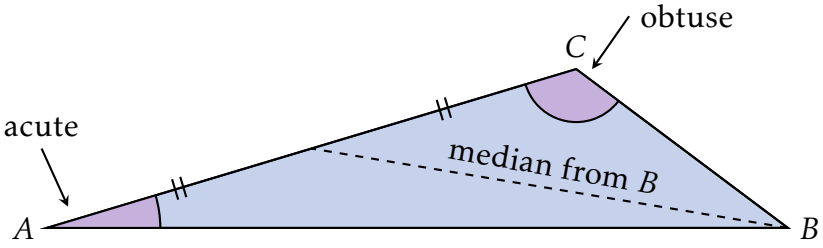
Given that we have all of these tools, we can use vectors to prove various geometric theorems!

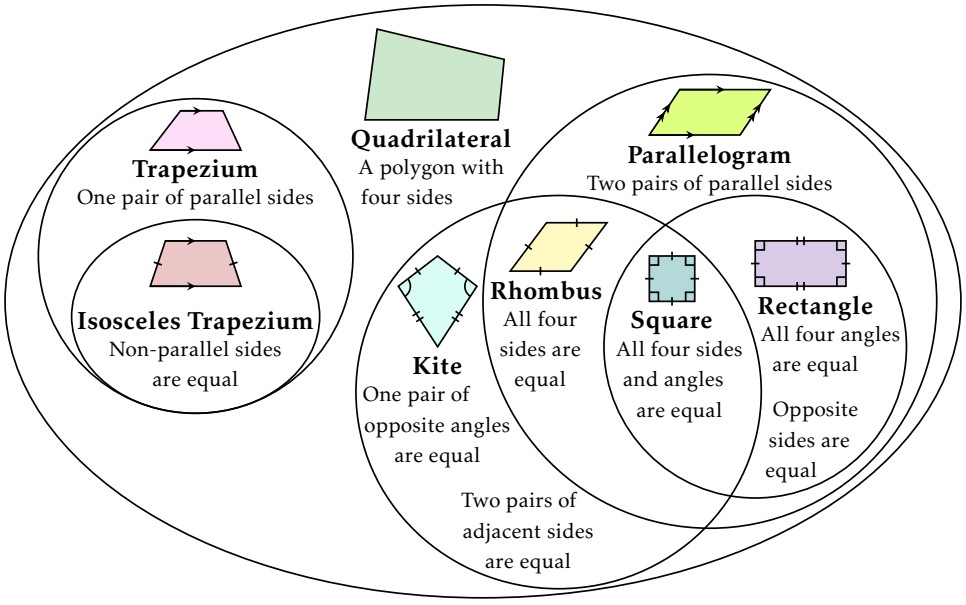
From above, you should know that

- lengths of sides are proved by using the magnitude of a vector
- perpendicularity is proved by showing the dot product of two vectors is equal to 0
- parallelism is proved by showing that one vector is a scalar multiple of another
- ratios on a line segment are proved using linear combinations of vectors
- various properties are proved by showing that multiple vector expressions lead to the same position vector

As a quick recap before we attempt some geometric proofs, the diagrams below contain some important geometric terms that you should familiarise yourself with.[3]







Finally, take a moment to go through this list of general vocabulary and ensure you are familiar with all these terms. If you do not understand any, make sure to ask your tutor for help!

Circles	Triangles	Quadrilaterals
• Radius	• Equilateral	• Square
• Diameter	• Isosceles	• Rectangle
• Circumference	• Scalene	• Trapezium
• Tangent	• Acute	• Parallelogram
• Arc	• Obtuse	• Rhombus
• Chord	• Similar	• Kite
• Sector	• Congruent	• Vertex
• Segment	• Median	• Edge
• Semicircle	• Hypotenuse	• Diagonal

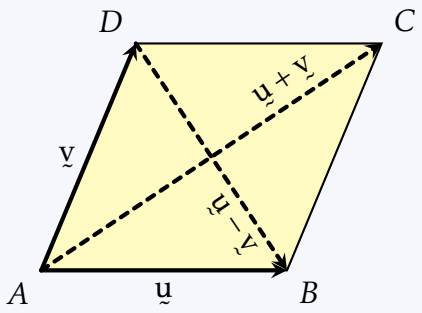
Example 4.1 Perpendicular Diagonals

Prove that the diagonals of a rhombus are perpendicular.

Solution:
Let $ABCD$ be a rhombus.

We want to show that $\overrightarrow{AC} \cdot \overrightarrow{DB} = 0$.

Define $\overrightarrow{AB} = \underset{\sim}{u}$ and $\overrightarrow{AD} = \underset{\sim}{v}$. Since $ABCD$ is a rhombus, $|\underset{\sim}{u}| = |\underset{\sim}{v}|$.



Then $\overrightarrow{AC} \cdot \overrightarrow{DB} = (\underset{\sim}{u}+\underset{\sim}{v}) \cdot (\underset{\sim}{u}-\underset{\sim}{v}) = \underset{\sim}{u} \cdot \underset{\sim}{u} - \underset{\sim}{u} \cdot \underset{\sim}{v} + \underset{\sim}{v} \cdot \underset{\sim}{u} - \underset{\sim}{v} \cdot \underset{\sim}{v} = \underset{\sim}{u} \cdot \underset{\sim}{u} - \underset{\sim}{v} \cdot \underset{\sim}{v}$

Now we use the the identity $\underset{\sim}{u} \cdot \underset{\sim}{u} = |\underset{\sim}{u}|^2$.

Therefore $\underset{\sim}{u} \cdot \underset{\sim}{u} - \underset{\sim}{v} \cdot \underset{\sim}{v} = |\underset{\sim}{u}|^2 - |\underset{\sim}{v}|^2 = |\underset{\sim}{u}|^2 - |\underset{\sim}{u}|^2 = 0$.

Since $\overrightarrow{AC} \cdot \overrightarrow{DB} = 0$, the diagonals of a rhombus are perpendicular. □

Challenge 4.1 Parallelogram Bisection

Prove that the diagonals of a parallelogram bisect each other.

Student Notes 4.1 Geometric Proofs

[3]Venn diagram inspired by Math Monks Quadrilateral Venn Diagram.

4.1 Collinearity

Consider the problem of checking whether three points lie on the same line. This could be done by using regular linear equations, but we could also attempt this question using vectors.

Challenge 4.2 Three Points

Let A, B and C be arbitrary points. Come up with a rule using vectors that allows us to test if the points A, B and C lie on the same line.

?

Challenge 4.3 Four Points

Consider the points $A(1,-2,3)$, $B(3,2,7)$, $C(5,6,11)$ and $D(-2,m,n)$, where $m,n \in \mathbb{R}$. Show that A, B and C lie on the same line using the rule from Challenge 4.2. Hence, find the values of m and n such that D also lies on this same line.

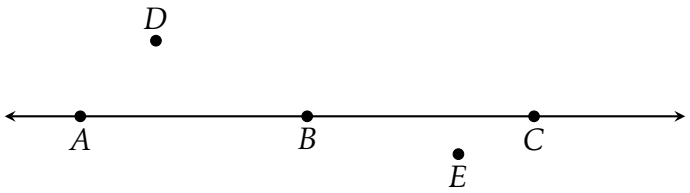
?

Consider how you may attempt Challenge 4.3 without using vectors. Does this alternative method seem easier or harder? What may be the benefits of the vector approach?

When working in two dimensions, checking whether points lie on a line is often done by comparing gradients. However, this approach quickly becomes cumbersome in three dimensions, where lines no longer have a single well-defined gradient. Vectors provide a unified method that works in any dimension and avoids unnecessary algebra.

Definition 4.1 Collinearity

A set of three or more points that lie on the same straight line are called **collinear**.



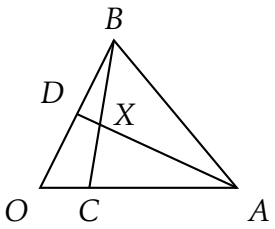
The key observation is that if three points lie on the same straight line, then the vectors connecting them must all point in the same (or opposite) direction. This idea allows us to replace geometric reasoning with a simple algebraic test involving scalar multiples.

Theorem 4.1 Proving Collinearity

Three points A, B and C are collinear if and only if $\overrightarrow{AB} = k\overrightarrow{AC}$ for some $k \in \mathbb{R}$.

And that marks the beginning into the world of vectors in Specialist Mathematics 3/4! In our next section, we explore how vectors can represent geometric objects such as lines and planes. Again, we highly recommend watching 3Blue1Brown's Essence of Linear Algebra series as a way to build visual intuition for these upcoming topics.

Question 4.1 (4 marks)
Consider a triangle $\triangle OAB$ where D is the midpoint of $\overline{OB}$ and $\overline{OA} = 4\overline{OC}$.



Let $\overrightarrow{OA} = \underset{\sim}{a}$, $\overrightarrow{OB} = \underset{\sim}{b}$, $\overrightarrow{CX} = s\overrightarrow{CB}$ and $\overrightarrow{DX} = t\overrightarrow{DA}$, where $s, t \in \mathbb{R}$.

a) Express $\overrightarrow{OX}$ in terms of

(i) $\underset{\sim}{a}$, $\underset{\sim}{b}$ and s. **1 mark**

(ii) $\underset{\sim}{a}$, $\underset{\sim}{b}$ and t. **1 mark**

b) Hence, find the values of s and t. **2 marks**

Question 4.2 (3 marks)
Prove that the midpoints of a quadrilateral form a parallelogram.

End of Question and Answer Book

Additional Answer Page: Please clearly number each question

Additional Answer Page: Please clearly number each question